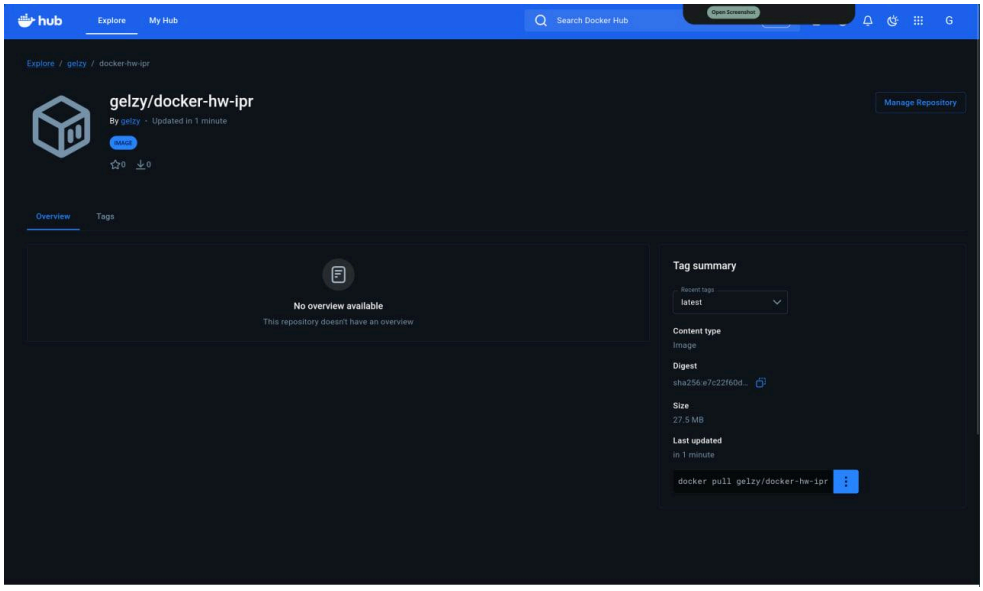
1.1

```
~/docker-hw v3.14.7 venv
docker build -t gelzy/docker-hw-ipr .
[+] Building 1.9s (7/7) FINISHED                                                
```

1.3

```
~/docker-hw v3.14.7 venv
docker push gelzy/docker-hw-ipr:latest
The push refers to repository [docker.io/gelzy/docker-hw-ipr]
e2de96513ba9: Pushed
6711e0c4f5a7: Pushed
df34abebe416: Pushed
70683d58fa5c: Pushed
3bf3671a4e6a: Pushed
6638b436592c: Pushed
1fdd0614a47e: Pushed
710ec83ca6cd: Pushed
52fe9fa75fd9: Pushed
b44956913298: Pushed
latest: digest: sha256:e7c22f60d73300bcf514ce9370a582b1f89a9ed5c6b6e30c980f5612098bcd4e size: 856
```

1.4



1.5

```
~/docker-hw v3.14.7 venv
docker image ls

IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest   c4d56c24da4f       22.2MB         6.03MB
gelzy/docker-hw-ipr:latest        e7c22f60d733        102MB         28.8MB    U
hello-world:latest                5e2309035332       25.9kB         9.49kB    U

~/docker-hw v3.14.7 venv
docker rmi -f e7c22f60d733 && docker pull gelzy/docker-hw-ipr:latest
Untagged: gelzy/docker-hw-ipr:latest
Deleted: sha256:e7c22f60d73300bcf514ce9370a582b1f89a9ed5c6b6e30c980f5612098bcd4e
latest: Pulling from gelzy/docker-hw-ipr
df34abebe416: Download complete
Digest: sha256:e7c22f60d73300bcf514ce9370a582b1f89a9ed5c6b6e30c980f5612098bcd4e
Status: Downloaded newer image for gelzy/docker-hw-ipr:latest
docker.io/gelzy/docker-hw-ipr:latest

~/docker-hw v3.14.7 venv
docker image ls

IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest   c4d56c24da4f       22.2MB         6.03MB
gelzy/docker-hw-ipr:latest        e7c22f60d733        102MB         28.8MB    U
hello-world:latest                5e2309035332       25.9kB         9.49kB    U

~/docker-hw v3.14.7 venv
```

2.1

```yaml
services:
    ▷Run Service
    frontend:
        image: gelzy/docker-hw-ipr:latest
        ports:
            - "8080:80"

    ▷Run Service
    api:
        image: containous/whoami

    ▷Run Service
    logger:
        image: busybox
        volumes:
        - ./logger.sh:/logger.sh
        - logger-data:/data
        command: ["sh", "/logger.sh"]

volumes:
    logger-data:
```

2.2

```
~/docker-hw  v3.14.7  venv
docker compose up -d
[+] up 4/4
 ✓ Volume docker-hw_logger-data   Created
 ✓ Container docker-hw-frontend-1 Running
 ✓ Container docker-hw-api-1      Running
 ✓ Container docker-hw-logger-1   Started

~/docker-hw  v3.14.7  venv
docker compose ps
NAME                IMAGE                          COMMAND                  SERVICE    CREATED          STATUS          PORTS
docker-hw-api-1     containous/whoami              "/whoami"                api        39 minutes ago   Up 39 minutes   80/tcp
docker-hw-frontend-1   gelzy/docker-hw-ipr:latest  "/docker-entrypoint.…"   frontend   39 minutes ago   Up 39 minutes   0.0.0.0:8080→80/tcp, [::]:8080→80/tcp
docker-hw-logger-1  busybox                        "sh /logger.sh"          logger     37 seconds ago   Up 26 seconds
```

2.3

```
~/ /docker-hw  v3.14.7 venv
docker compose exec frontend sh

/ # curl http://api
Hostname: 080c9a32c186
IP: 127.0.0.1
IP: ::1
IP: 172.18.0.3
RemoteAddr: 172.18.0.2:51264
GET / HTTP/1.1
Host: api
User-Agent: curl/8.22.0
Accept: */*

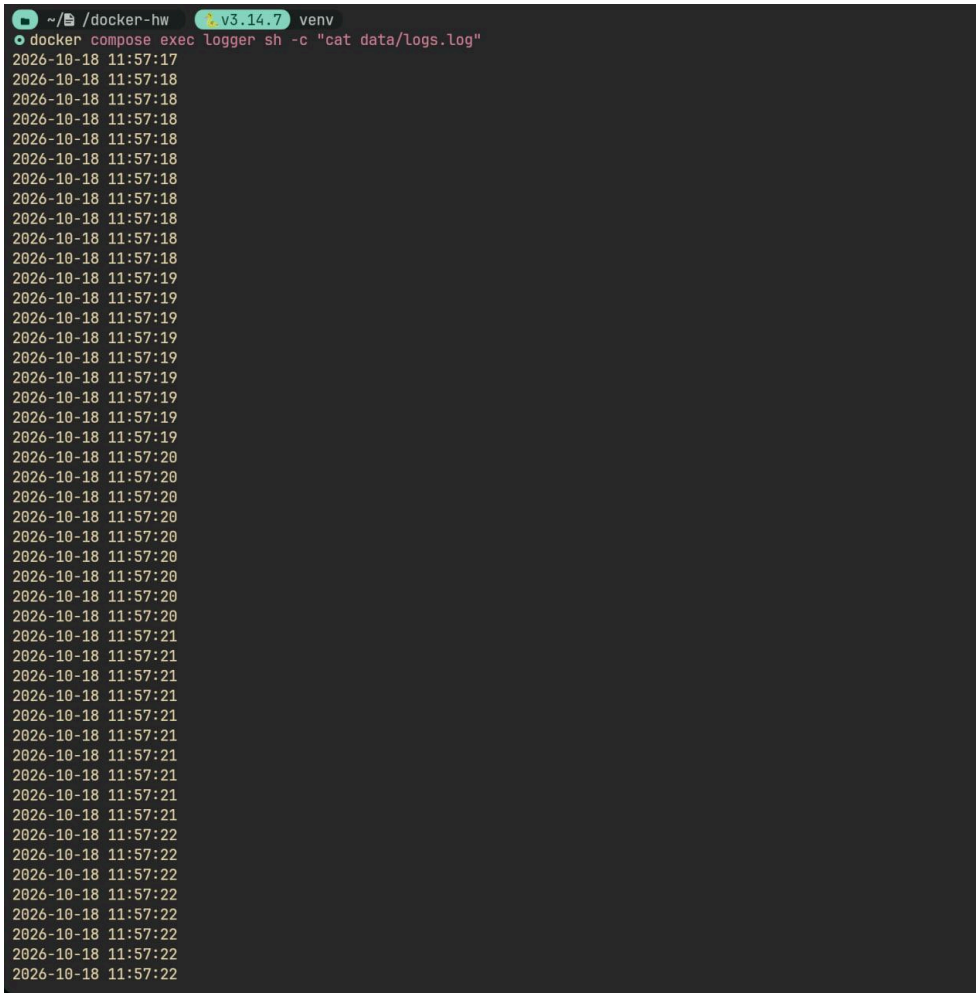
/ #
```

2.4

```
~/ /docker-hw  v3.14.7 venv
docker volume ls | grep logger
local     docker-hw_logger-data

~/ /docker-hw  v3.14.7 venv
```

2.5


~/ /docker-hw
v3.14.7
venv
docker compose exec logger sh -c "cat data/logs.log"
2026-10-18 11:57:17
2026-10-18 11:57:18
2026-10-18 11:57:18
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2026-10-18 11:57:22

3.1 + 3.2

```
~/ /docker-hw  v3.14.7  venv
docker compose exec logger sh -c "wc -l data/logs.log"
444 data/logs.log

~/ /docker-hw  v3.14.7  venv
docker compose down
[+] down 4/4
 ✔ Container docker-hw-logger-1   Removed
 ✔ Container docker-hw-frontend-1 Removed
 ✔ Container docker-hw-api-1      Removed
 ✔ Network docker-hw_default      Removed

~/ /docker-hw  v3.14.7  venv
docker compose up -d
[+] up 4/4
 ✔ Network docker-hw_default      Created
 ✔ Container docker-hw-logger-1   Started
 ✔ Container docker-hw-api-1      Started
 ✔ Container docker-hw-frontend-1 Started

~/ /docker-hw  v3.14.7  venv
docker compose exec logger sh -c "wc -l data/logs.log"
1117 data/logs.log
```

3.3

```
~/ /docker-hw  v3.14.7  venv
docker compose down -v
[+] down 5/5
 ✔ Container docker-hw-logger-1   Removed
 ✔ Container docker-hw-frontend-1 Removed
 ✔ Container docker-hw-api-1      Removed
 ✔ Network docker-hw_default      Removed
 ✔ Volume docker-hw_logger-data   Removed

~/ /docker-hw  v3.14.7  venv
docker compose up -d
[+] up 5/5
 ✔ Network docker-hw_default      Created
 ✔ Volume docker-hw_logger-data   Created
 ✔ Container docker-hw-logger-1   Started
 ✔ Container docker-hw-api-1      Started
 ✔ Container docker-hw-frontend-1 Started

~/ /docker-hw  v3.14.7  venv
docker compose exec logger sh -c "wc -l /data/logs.log"
39 /data/logs.log
```